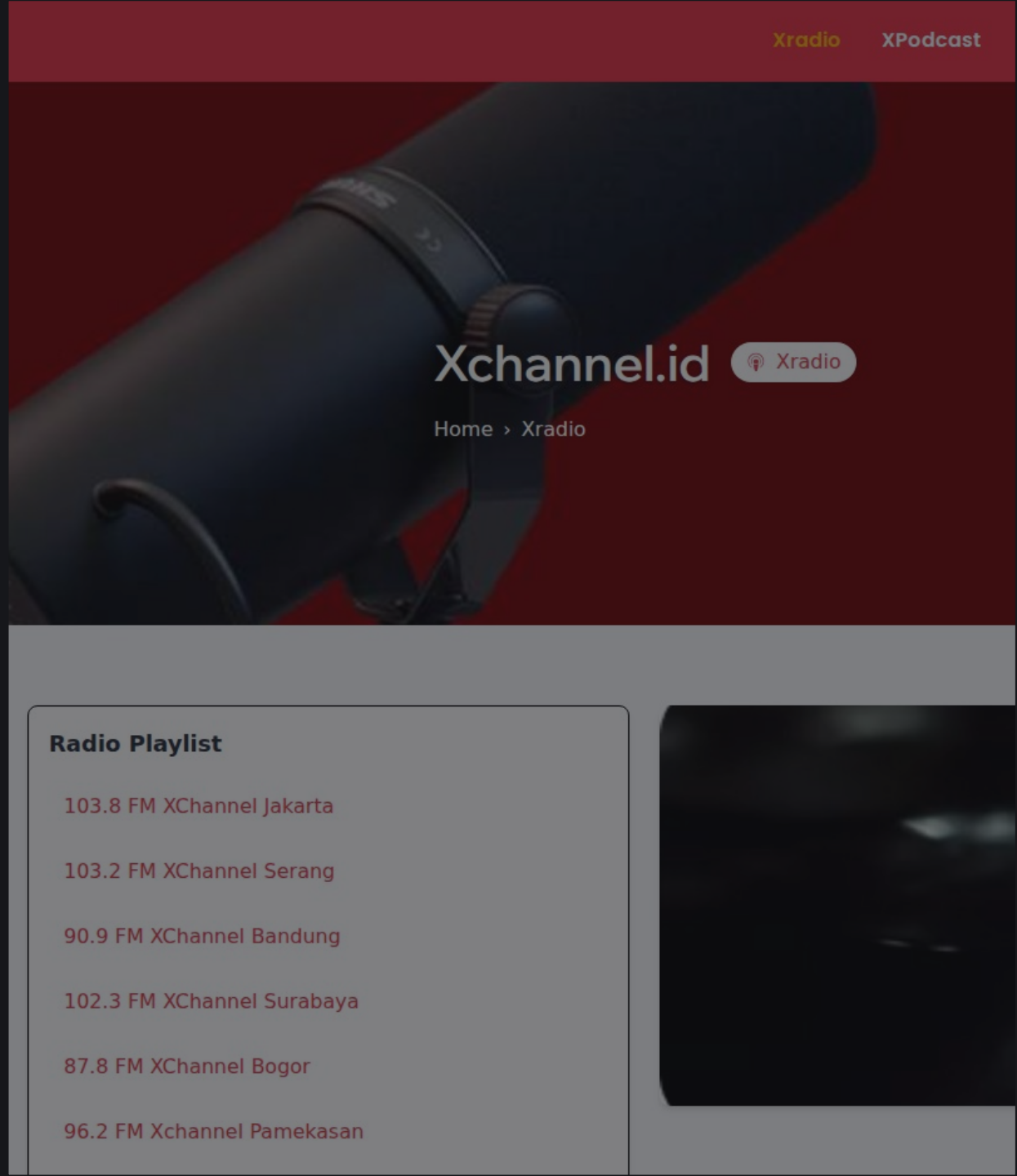
XRADIO · DASHBOARD

XChannel

The broadcast schedule for local radio in several cities, and the system behind it

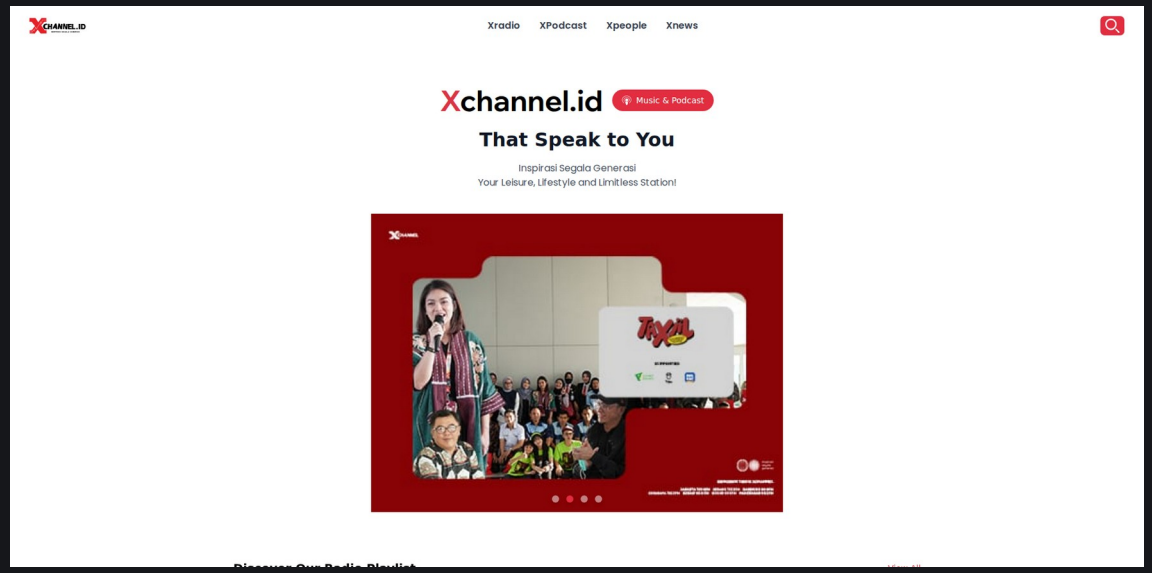
Enrico Saputra · Backend developer, Laravel dashboard

Client code and data are private. The demo rebuilds the radio part of the dashboard on the real data model, with made-up data.

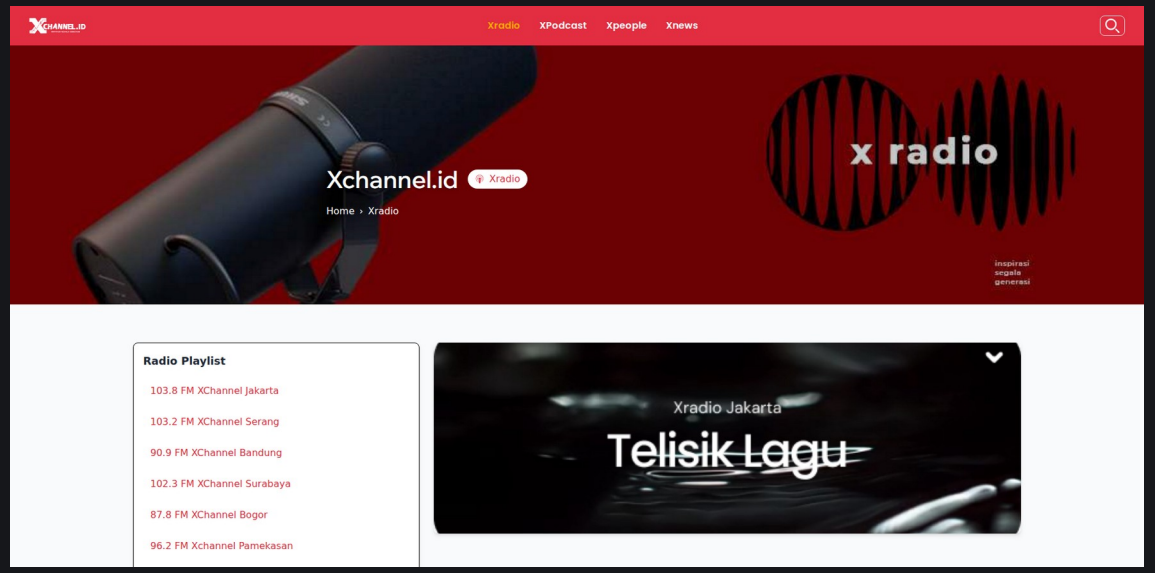


THE REAL PRODUCT

xchannel.id, the side listeners see



Home: radio, podcasts, people and news



Xradio: stations per city and the show that is playing

The stations, shows, hosts and schedule on the Xradio page are managed in the Laravel dashboard. That is the part this demo rebuilds.

Screenshots of xchannel.id, October 2026

MY PART

One dashboard, many stations

The XChannel app plays each station's stream and shows what is on air. The broadcast team runs all of it from the dashboard.

What I built

- Stations: city, frequency, stream URL and status
- Shows and hosts
- Each station's weekly schedule, which the app reads
- Podcasts, chart, video, news: same dashboard, outside this demo

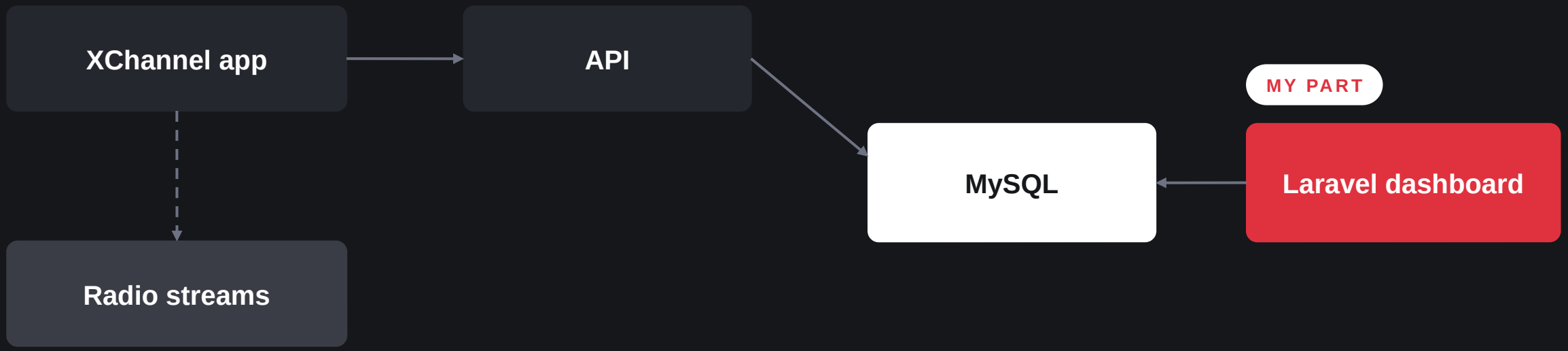
8 stations on xchannel.id today

7 → 1 per-city schedule tables become one

24/7 empty hours play automated music

ARCHITECTURE

One database, two readers



The dashboard writes

Stations, shows, hosts and the schedule.

The app reads

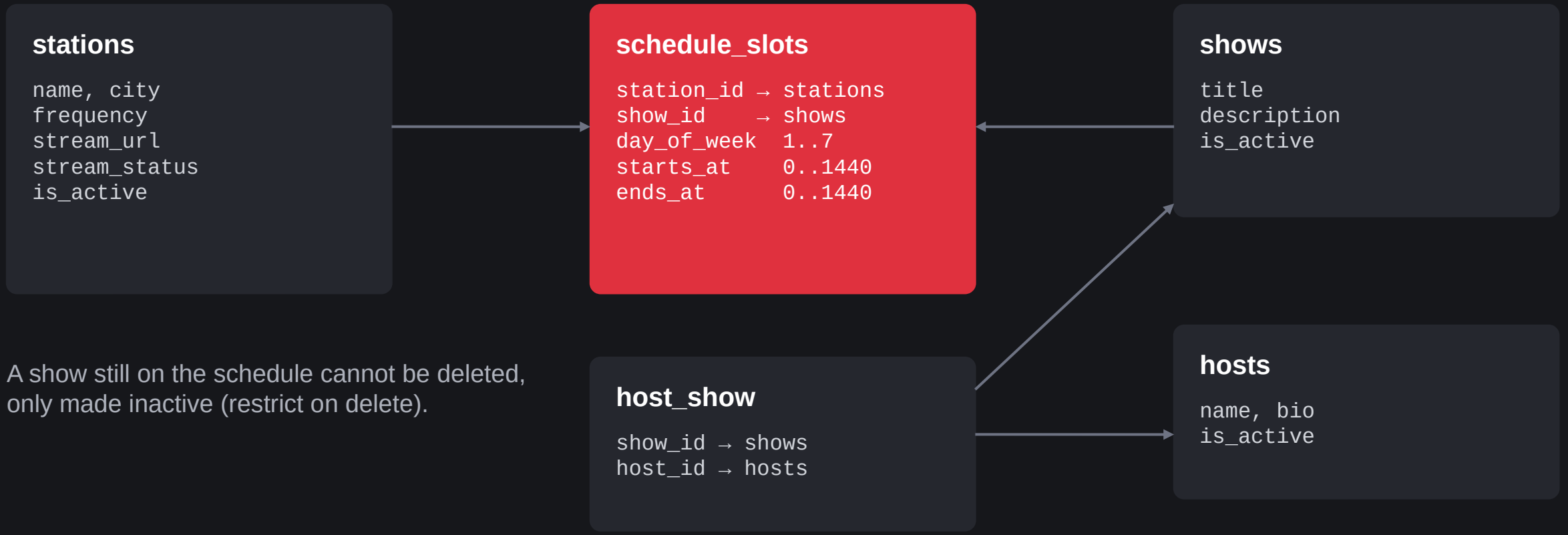
"What is on air now?" through its API.

Rules on save

An overlapping schedule never reaches a listener.

DATA MODEL

Four tables and one pivot



A show still on the schedule cannot be deleted, only made inactive (restrict on delete).

PRODUCTION VS DEMO

From a table per city to one table

Production

7 TABLES

```
radio.api_jadwal = 'jadwal_radio_woman'

jdwl_radio_bandung
jdwl_radio_surabaya
jdwl_radio_suramadu  ...
  judul_siaran, hari_siaran,
  waktu_siaran, penyiar (text)
```

A new station means a new table. One name in api_jadwal does not match the real table.

Demo

1 TABLE

```
schedule_slots
  station_id  -> stations
  show_id     -> shows
  day_of_week, starts_at, ends_at

host_show (show_id, host_id)
```

A new station is one row. Foreign keys keep it consistent.

No-downtime migration: fill the new table from the old ones, keep a view per city for the old API, move readers one by one.

TIME

Time as numbers, not text

In production

`senin-jumat`	`Senin- Jumat`
`Monday-Friday`	`6:00 AM - 10:00 AM`
`08.00-09.00`	`20:00-24:00`

In the demo

450

minutes = 07:30

1440

minutes = 24:00

day_of_week 1-7 (ISO), one row per day. The form takes HH:MM only.

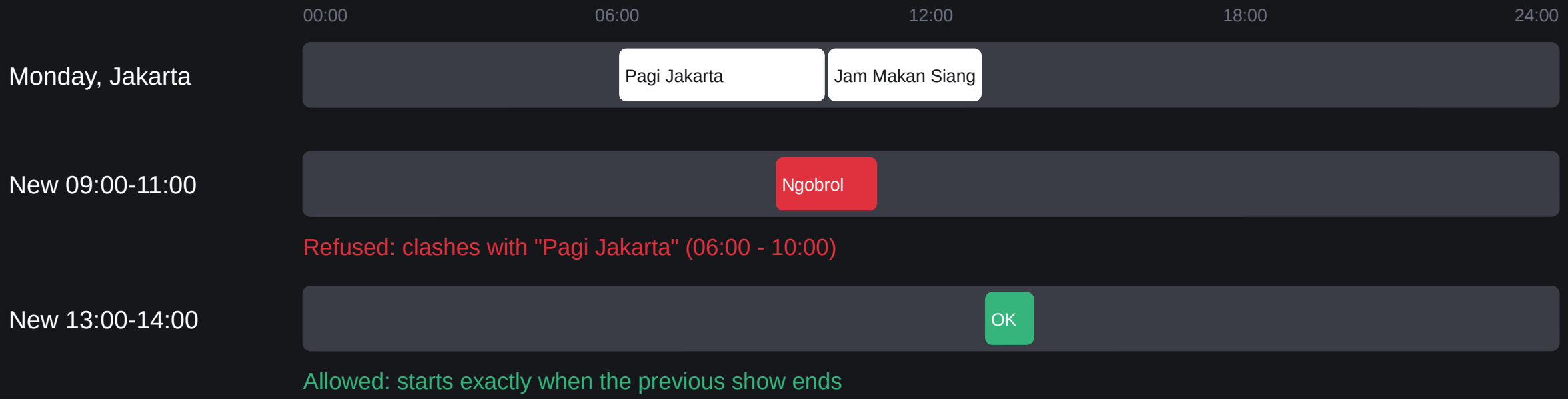
Two slots overlap when each starts before the other ends

```
starts_at < :end  AND  ends_at > :start
```

Touching (10:00 then 10:00) is fine.

OVERLAPS

The overlap rule on save



A clear message

Day, time and the show in the way.

All days or none

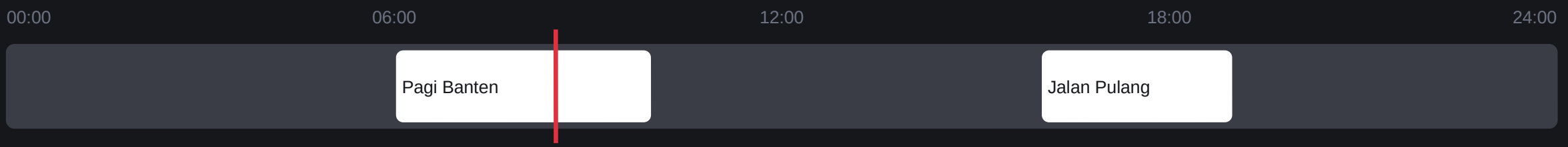
Monday to Friday in one transaction.

Past midnight

Two slots: until 24:00, then from 00:00.

RIGHT NOW

On air, up next, and empty hours



XChannel Banten, Monday 08:30. Red line = now; grey = automated music.

On air

Today's slot with start <= now < end. One indexed query.

Up next

The first slot after now, rolling into the next days for up to a week.

Empty hours

Worked out from the schedule, not stored. Airtime per station out of 168 hours.

DECISIONS

Decisions worth talking about

Minutes, not TIME

24:00 is not a valid TIME. Integers behave the same on MySQL, PostgreSQL and SQLite.

Split at midnight

One slot, one day, so every query stays simple.

Co-host airtime

Each host counts the full slot: both of them work it.

Deactivate, not delete

Anything still in use can only be made inactive.

TECH STACK

Tech stack and why



Laravel 10

As in production. Forms, validation and CRUD are quick; the rules stay in one service.



MySQL

The production database. Tests on SQLite, online demo on PostgreSQL: the code is not tied to one.



Blade + Tailwind

The schedule timelines are plain HTML and CSS, no JavaScript.



PHPUnit

19 feature tests for the schedule rules, language and demo reset.

CODE

Code that reads easily

```
public function clash(int $stationId, Day $day,
    int $start, int $end, ?int $ignoreId = null)
{
    // Two ranges overlap when each one starts
    // before the other ends. Touching is fine.
    return ScheduleSlot::with('show')
        ->where('station_id', $stationId)
        ->where('day_of_week', $day->value)
        ->where('starts_at', '<', $end)
        ->where('ends_at', '>', $start)
        ->when($ignoreId, fn ($q) =>
            $q->whereKeyNot($ignoreId))
        ->first();
}
```

Rules in one service

Clash, schedule, move, on air, up next, gaps.

Enums, not numbers

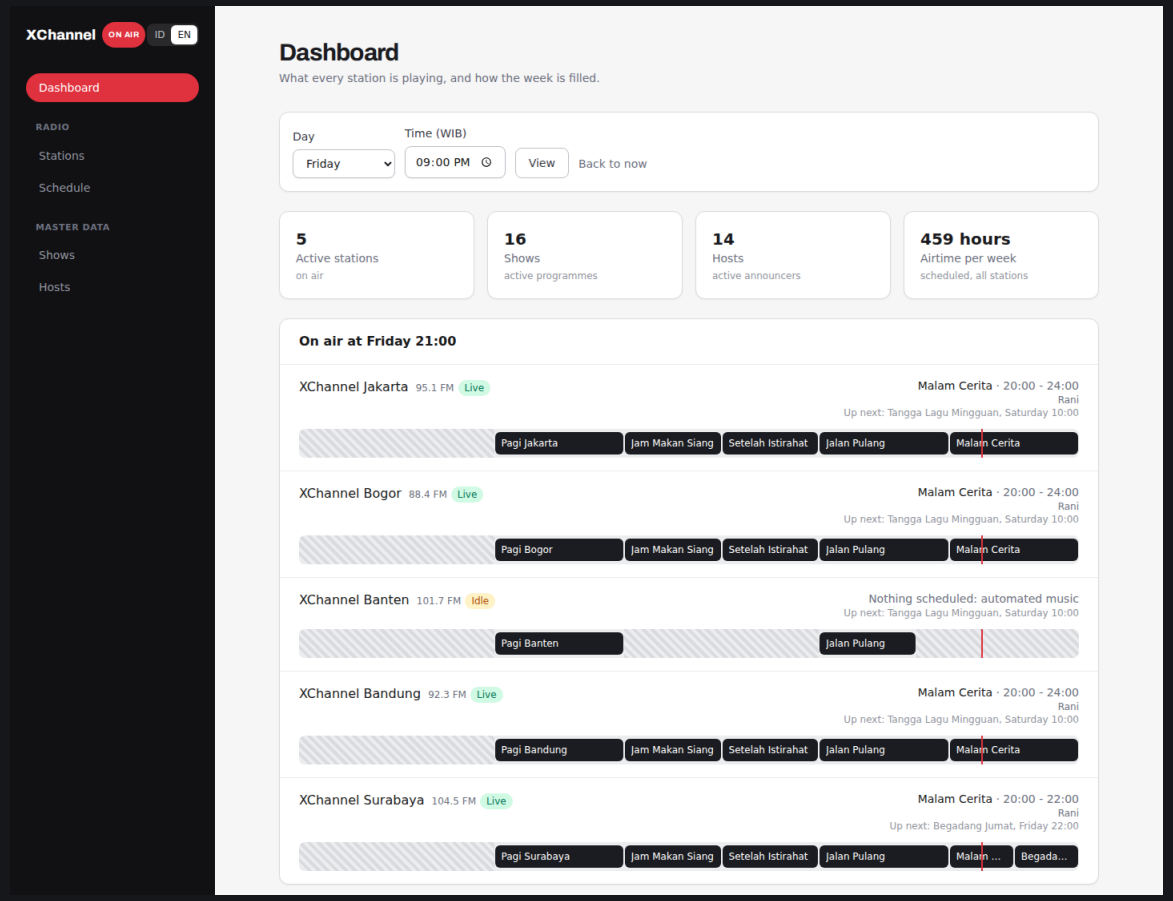
Day (ISO 1-7), StreamStatus.

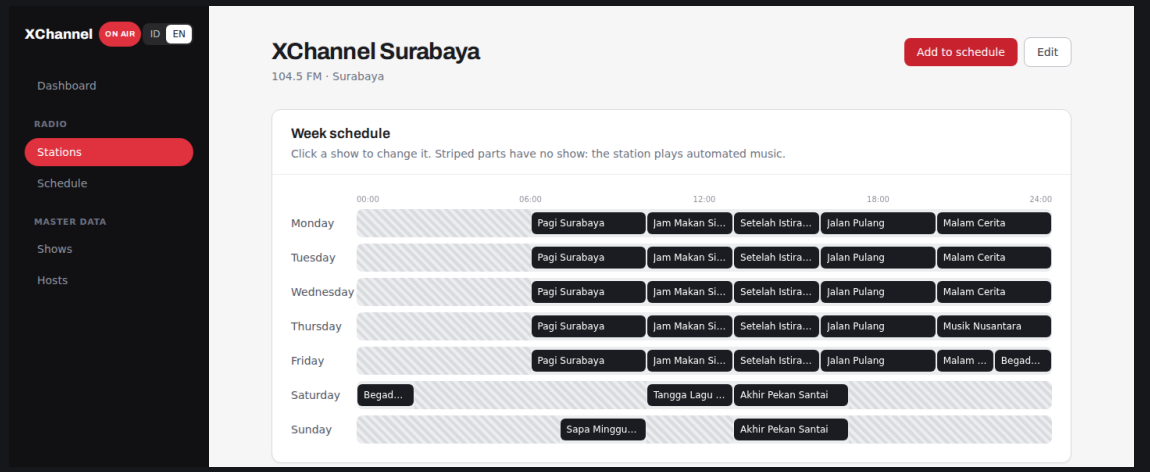
Thin controllers

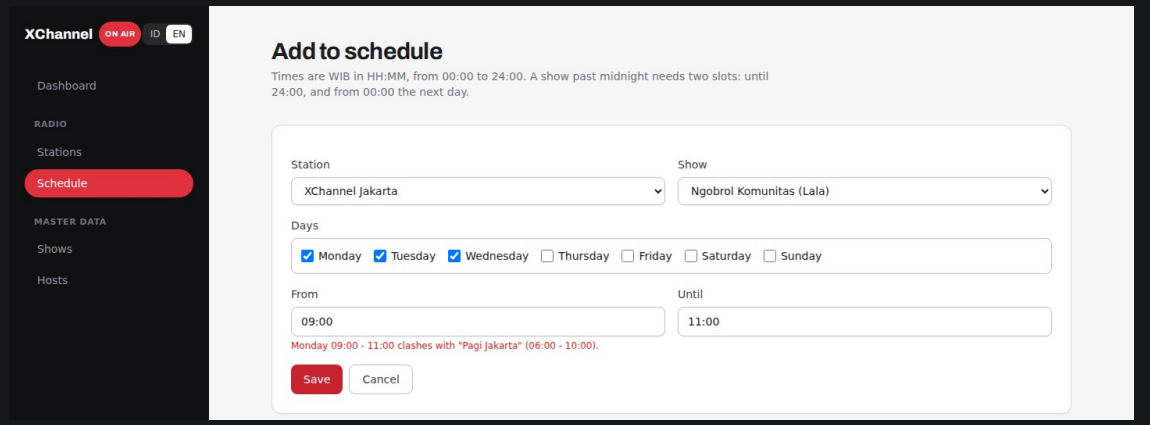
FormRequests validate, Clock parses times.

DEMO

The demo app







Dashboard: on air and airtime

NEXT

What I would improve

01 **Move to one table** — A view per city for the old API, readers moved over gradually.

02 **A time zone per station** — For stations in WITA or WIT.

03 **One-off changes** — Holidays and special broadcasts without editing the weekly pattern.

04 **Stream status from the stream** — live / idle / offline from a health check.

TRY IT

Try it yourself

Live demo	https://xchannel-dashboard-demo.vercel.app
Slides (PDF)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pdf
Slides (PPTX)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pptx
Real product	https://xchannel.id/radio

The demo code is in a private GitHub repo; access is given during the interview. The dashboard speaks Bahasa Indonesia and English, and its data is restored every 3 hours.